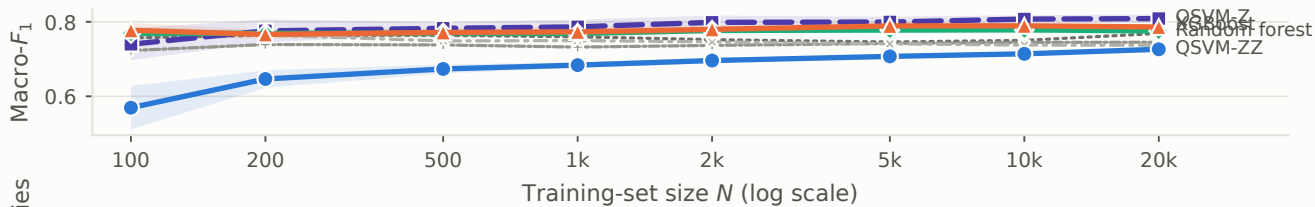
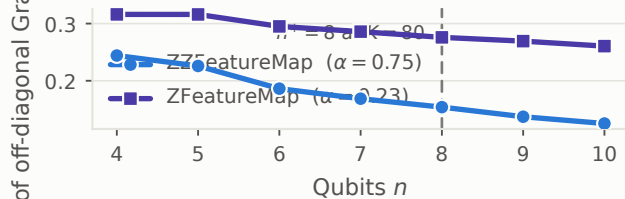


Widening the circuit destroys the quantum kernel, and the kernel geometry says why

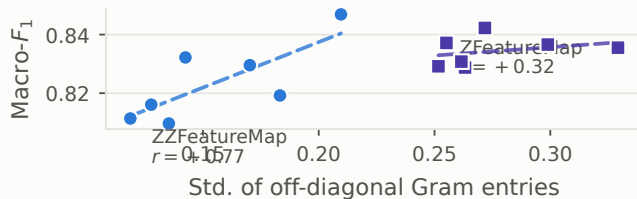
(a) $K = 80$, so the rule gives $n^* = 8$



(b) The kernel concentrates as n grows



(c) Dispersion predicts F_1



Legend: QSVM-ZZ (blue circle), QSVM-Z (purple square), XGBoost (orange triangle), Random forest (green diamond), SVM-RBF (dotted line), SVM-poly2 (dashed line), SVM-linear (dash-dot line), Holm-adj. $p < 0.05$ (light blue shaded region).